

Nguyen Manh Hiep

Software Engineer

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SUMMARY

I am a full-stack engineer looking for a role where I can own a product end to end. At ChimAI I delivered two production platforms: a notarial document digitization system on FastAPI, PostgreSQL and a Celery queue driving AI OCR, and the FastAPI backend, React dashboards and infrastructure of a sales-automation SaaS. I design REST APIs and the relational model behind them, build the Next.js and TypeScript interfaces that consume them, and integrate LLMs where they do real work inside a business process. I am drawn to the parts of a system that must stay correct under load: task queues, schedulers, idempotent webhooks and deduplication. I write code covered by real, non-mocked integration tests, and I ship what I build with Docker, Nginx and GitHub Actions.

EXPERIENCE

Notarex - notarial document digitization platform

03/2026 - 09/2026

Software Engineer, ChimAI JSC (Full-stack + AI document processing + Infrastructure):

- **Description:** A platform that digitizes notarial paperwork for notary offices in Vietnam, covering configurable form templates, AI-driven OCR extraction and a full document edit, save and download cycle.
- **Technologies & Tools:** Next.js 14, TypeScript, Ant Design, Zustand, TanStack Query, FastAPI (Python 3.11), SQLAlchemy 2.0 async, PostgreSQL 16, Celery, Redis, MinIO, OnlyOffice, docxtempl, Nginx, Docker, GitHub Actions
- **Key Responsibilities & Contributions:**
 - **End-to-end Feature Development:** Built the digitization flow from configurable form templates through AI OCR extraction to a JWT-protected OnlyOffice edit-save-download cycle, with automated DOCX generation via docxtempl.
 - **Authorization Design:** Designed a three-role model (admin, director, employee) scoped per notary office, isolating each office's data from the others.
 - **Background Processing:** Built a Celery and Redis task queue on a ZSET pending queue with an atomic active-counter, plus heartbeat reconciliation that recovers zombie jobs, keeping OCR stable at scale.
 - **Infrastructure & DevOps:** Set up S3-compatible storage on MinIO, containerized the system with Docker behind Nginx, and built CI/CD with GitHub Actions.
 - **Customer Requirements & Delivery:** Collected requirements directly with notary offices and turned them into technical solutions, then scoped each release in sprints against a shared task board, in a 2 to 3 person team with no dedicated PM.

EazySale - sales lead generation and outreach SaaS

03/2026 - 08/2026

Software Engineer, ChimAI JSC (Backend + React dashboards + Infrastructure):

- **Description:** A SaaS for sales lead management and automated customer outreach over Zalo and email. I owned the FastAPI backend, three React dashboards and the infrastructure.
- **Technologies & Tools:** React 18, TypeScript, Ant Design, Material-UI, Zustand, FastAPI (Python 3.11), SQLAlchemy, Alembic, PostgreSQL 15, APScheduler, MinIO, Zalo OA API, SendGrid, Nginx, Docker, GitHub Actions, GHCR
- **Key Responsibilities & Contributions:**
 - **Zalo OA Integration:** Built the Zalo OA ZNS API channel with template management, HMAC-SHA256 webhook verification and delivery metrics, plus a mock provider with a tunable failure rate for testing without spending quota.
 - **Scheduled Campaign Delivery:** Scheduled Excel-driven campaigns with APScheduler date triggers backed by a 60-second database poll that recovers jobs lost to a restart, with SELECT FOR UPDATE NOWAIT so a message is sent once even when both paths fire.
 - **Email Campaigns & Event Tracking:** Built SendGrid campaigns whose public event webhook is made idempotent by in-batch deduplication and a status-priority guard inside the UPDATE, so out-of-order and repeated retries cannot roll a delivery state backwards.
 - **Lead Data Quality:** Built a duplicate detector that checks uploaded leads against each other and against existing records, normalizing phone-number variants, and converted Excel date serials to DD/MM/YYYY before building Zalo payloads.
 - **Architecture & Delivery:** Designed a single-domain, path-routed deployment (login, admin, customer app, API) on Nginx with Let's Encrypt TLS, RBAC with per-user data isolation across 129 REST endpoints, and a GitHub Actions pipeline building four images to GHCR and deploying to the VPS with migrations and zero downtime.

PROJECTS

SenAI - real-time Vietnamese-English meeting translation

2026

Frontend, Backend and AI Integration Owner (Real-time systems + LLM orchestration + Infrastructure):

- **Description:** A web platform that translates live meetings between Vietnamese and English, built by a 5-person team for the Vietnam AI Innovation Challenge 2026, where it won Third Prize.
- **Technologies & Tools:** Next.js 16, React 19, TypeScript, Tailwind v4, FastAPI, PostgreSQL 16 + pgvector, LiveKit, WebSocket, vLLM, Docker, Nginx, GitHub Actions, systemd, Amazon SES
- **Key Responsibilities & Contributions:**
 - **AI/LLM Integration:** Built an extensible model router for ASR, fast translation, refined translation, TTS and LLM that switches provider at runtime between self-hosted vLLM and commercial APIs, with fallback handling on failure.
 - **Translation Pipeline:** Ran an independent ASR and translation pipeline per microphone to separate audio by speaker and stream translated subtitles, pairing fast translation with LLM refinement against a contextual glossary.
 - **Real-time Meeting System:** Built LiveKit (WebRTC) rooms with camera, microphone, screen sharing, translated two-way chat, waiting-room approval, host controls, QR invitations and configuration updates applied mid-meeting.
 - **Backend & Data Layer:** Built FastAPI services for meeting configuration, token generation, WebSocket authentication, model routing and bilingual transcript storage, and migrated the data layer from SQLite to PostgreSQL 16 + pgvector without changing public function signatures.
 - **Frontend & Delivery:** Shipped the Next.js application across 7 feature areas with its own design system, Vietnamese-English i18n, dark mode and PWA support, deployed by GitHub Actions to a VPS behind Nginx and systemd, and covered by 37 UAT suites containing 147 real, non-mocked tests.

nguyenmanhhiiep.info - personal portfolio

2026

Self-built bilingual Vietnamese-English portfolio hosting my work and both CV versions, deployed and maintained by me.

EDUCATION

Hanoi University of Science and Technology (HUST)

2021 - 2026 (expected)

Computer Science

HONORS & AWARDS

Hackathon Third Prize at Vietnam AI Innovation Challenge

2026

TECHNICAL SKILLS

AI-assisted development	Claude Code (agents, skills, hooks), Cursor, ChatGPT, Gemini, GitHub Copilot, MCP servers, prompt & context engineering, spec-driven AI workflows, RAG, LLM orchestration, model routing with fallback, vLLM self-hosting
Frontend	React 19/18, Next.js 16/14, TypeScript, Tailwind v4, Ant Design, Material-UI, Zustand, TanStack Query, PWA, i18n
Backend	FastAPI, Python, Pydantic, RESTful API design, Node.js, NestJS / ExpressJS, Spring Boot, Java, WebSocket, WebRTC (LiveKit), Celery, APScheduler, Redis caching & queues, JWT / OAuth, RBAC, multi-tenant data isolation, third-party API & webhook integration (Zalo OA, SendGrid)
Databases	PostgreSQL 16/15 (+ pgvector), MySQL, MariaDB, SQL query optimization & indexing, SQLAlchemy 2.0 async, Alembic, schema migrations, vector search & embeddings, MinIO (S3)
DevOps & Cloud	Docker & Docker Compose, Nginx, systemd, TLS / Let's Encrypt, VPS deployment, Git & GitHub workflows, GitHub Actions CI/CD, GHCR, Cloudflare, Amazon SES
Engineering practice	Systems thinking, system design, software architecture, SOLID, OOP, design patterns, code review, idempotency & concurrency control, web application security, real (non-mocked) integration & security testing, technical documentation (English), requirements analysis, SRS / Use Case / ERD
Soft Skills	Critical thinking, technical research, cross-functional teamwork, quick adaptability to new technologies